EDA Stress App

Biomedical Wearable Technologies for Healthcare and Wellbeing

Academic Year 2025/2026



Contributors

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«Technological environments become more demanding placing human operators in increasingly complex tasks… There is a need to assess the perceived cognitive load to avoid exceeding the operator's capabilities»

"cognitive load is a multifaceted construct that rarely occurs in isolation, it's hard to infer cognition from sudomotor and cardiac manifestations when analyzed individually…autonomic responses concern sudomotor control, cardiac control or both in different proportions so they are mirrored in EDA or HR or both in different proportions"



sensors

MDPI

Article

Detection of Cognitive Load Modulation by EDA and HRV

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Abstract: Electrodermal activity (EDA) and heart rate variability (HRV) offer opportunities to grasp critical manifestations of the nervous autonomic system using low-intrusive sensing tools. A key question relies on the capacity to adequately process EDA and HRV signals to extract cognitive load markers, a multifaceted construct with intricate neural networks functioning, where emotions interfere with cognition. Here, 34 participants (20 males, 19.2 ± 1.3 years) were exposed to two-back mental tasking and watching emotionally charged images while recording EDA and HRV. HRV signals were processed using variable frequency complex demodulation (VFCDM) and wavelet packet transform (WPT) to provide high- and low-frequency (HF and LF) markers. Three methods were used to extract EDA indices: VFCDM (EDA_{TVSYMP}), WPT (EDA_{WPT}), and convex-optimization (EDA_{CVX}). Cognitive load and emotion epochs were distinguished by significant differences in NASA-TLX scores, mental fatigue, and stress, on the one hand; and by EDA_{CVX} and, remarkably, EDA_{TVSYMP} and $HF\text{-}HRV_{VFCDM}$ on the other hand. A linear mixed-effects model and stepwise backward selection procedure showed that these two markers were main predictors of the NASA-TLX score (cognitive load). The individual perception of cognitive load was finally discriminated by k-means clustering, showing three profiles of autonomic responses relying, respectively, on EDA_{TVSYMP}, $HF\text{-}HRV_{VFCDM}$, or a mix of these two markers. The existence of EDA-, HRV-, and EDA/HRV-derived profiles might explain why previous attempts that have predominantly employed a single biosignal often remained unconclusive in evaluating the perceived cognitive load, thereby demonstrating the added value of the present approach to monitor mental-related workload in human operators.

Keywords: cognitive load; EDA; HRV; 2-back; NASA-TLX

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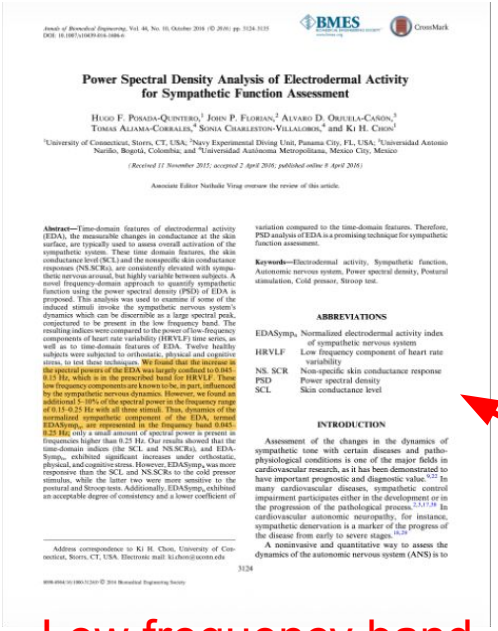
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1. Introduction

Human monitoring during a variety of mental tasks has become popular for exploring human performance, well-being, and safety in different contexts. As technological environments become more demanding, and due to the inherent risk of placing human operators in increasingly complex tasks, there is a need to assess the perceived cognitive load to avoid exceeding the user's abilities. In this context, neurophysiological research has provided a growing appreciation that a tight coupling exists between cortical activities in high-level cerebral assemblies sustaining cognitive functions and the activity of the autonomous nervous system, which is reflected in the control of end-effectors, e.g., the sudoral glands and the sinus node. This central and autonomous interplay offers a good opportunity to



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Power Spectral Density Analysis of Electrodermal Activity for Sympathetic Function Assessment

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Abstract—Time-domain features of electrodermal activity (EDA), the measurable changes in conductance at the skin surface, are typically used to assess overall activation of the sympathetic system. These time domain features, the skin conductance level (SCL) and the nonspecific skin conductance responses (NS.SCRs), are consistently elevated with sympathetic nervous arousal, but highly variable between subjects. A novel frequency-domain approach to quantify sympathetic function using the power spectral density (PSD) of EDA is proposed. This analysis was used to examine if some of the induced stimuli invoke the sympathetic nervous system's dynamics which can be discernible as a large spectral peak, conjectured to be present in the low frequency band. The resulting indices were compared to the power of low-frequency components of heart rate variability (HRVLF) time series, as well as to time-domain features of EDA. Twelve healthy subjects were subjected to orthostatic, physical and cognitive stress, to test these techniques. We found that the increase in the spectral powers of the EDA was largely confined to 0.045–0.25 Hz, which is in the prescribed band for HRVLF. These low frequency components are known to be, in part, influenced by the sympathetic nervous dynamics. However, we found an additional 5–10% of the spectral power in the frequency range of 0.15–0.25 Hz with all three stimuli. Thus, dynamics of the normalized sympathetic component of the EDA, termed $EDASymp_n$, are represented in the frequency band 0.045–0.25 Hz; only a small amount of spectral power is present in frequencies higher than 0.25 Hz. Our results showed that the time-domain indices (the SCL and NS.SCRs), and $EDASymp_n$, exhibited significant increases under orthostatic, physical, and cognitive stress. However, $EDASymp_n$ was more responsive than the SCL and NS.SCRs to the cold pressor stimulus, while the latter two were more sensitive to the postural and Stroop tests. Additionally, $EDASymp_n$ exhibited an acceptable degree of consistency and a lower coefficient of variation compared to the time-domain features. Therefore, PSD analysis of EDA is a promising technique for sympathetic function assessment.

Keywords—Electrodermal activity, Sympathetic function, Autonomic nervous system, Power spectral density, Postural stimulation, Cold pressor, Stroop test.

ABBREVIATIONS

$EDASymp_n$	Normalized electrodermal activity index of sympathetic nervous system
HRVLF	Low frequency component of heart rate variability
NS. SCR	Non-specific skin conductance response
PSD	Power spectral density
SCL	Skin conductance level

INTRODUCTION

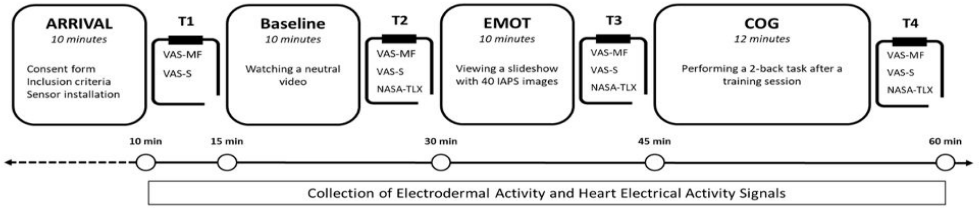
Assessment of the changes in the dynamics of sympathetic tone with certain diseases and pathophysiological conditions is one of the major fields in cardiovascular research, as it has been demonstrated to have important prognostic and diagnostic value.[9,22] In many cardiovascular diseases, sympathetic control impairment participates either in the development or in the progression of the pathological process.[2,3,17,38] In cardiovascular autonomic neuropathy, for instance, sympathetic denervation is a marker of the progress of the disease from early to severe stages.[16,29]

A noninvasive and quantitative way to assess the dynamics of the autonomic nervous system (ANS) is to

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Low frequency band influenced by sympathetic nervous dynamics:
0.045 – 0.25 Hz



3.2. *Physiological Indices*

3.2.1. Electrodermal Activity (EDA)

All indices derived from phasic EDA increased during COG when compared to Baseline and EMOT (Table 2), thus pointing to a heightened EDA response especially in COG condition.

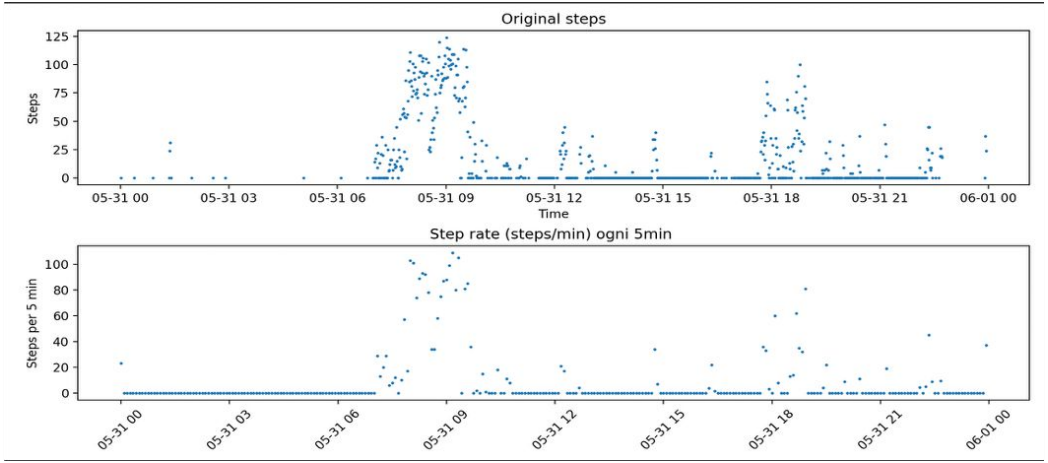
Table 2. Comparisons of physiological indices extracted from EDA and HRV.

Metrics	Units	Baseline	EMOT	COG
EDA_{CVX}	n.u	1.05 ± 0.20	1.05 ± 0.20	1.26 ± 0.28 ***,###
EDA_{TVSYMP}	n.u	0.80 ± 0.13	0.82 ± 0.12	1.04 ± 0.10 ***,###
EDA_{WPT3}	n.u	0.94 ± 0.14	0.90 ±0.17	1.04 ± 0.11 **,###
RMSSD	ms^2	45 ± 16	48 ± 18	48 ± 18
LF_{DFT}	ms^2	516 ± 308	693 ± 466	689 ± 468
HF_{DFT}	ms^2	310 ± 269	371 ± 284	329 ± 261
LF/HF_{DFT}	n.u	2.5 ± 2.2	2.5 ± 1.7	2.74 ± 1.65
LF_{VFCDM}	n.u	1.24 ± 0.04	1.22 ± 0.04	1.24 ± 0.03
HF_{VFCDM}	n.u	1.28 ± 0.05	1.28 ± 0.06	1.24 ± 0.04 **,##
LF_{WPT3}	n.u	1.21 ± 0.05	1.20 ± 0.05	1.22 ± 0.03
HF_{WPT3}	n.u	1.27 ± 0.05	1.27 ± 0.05	1.23 ± 0.5 *,#

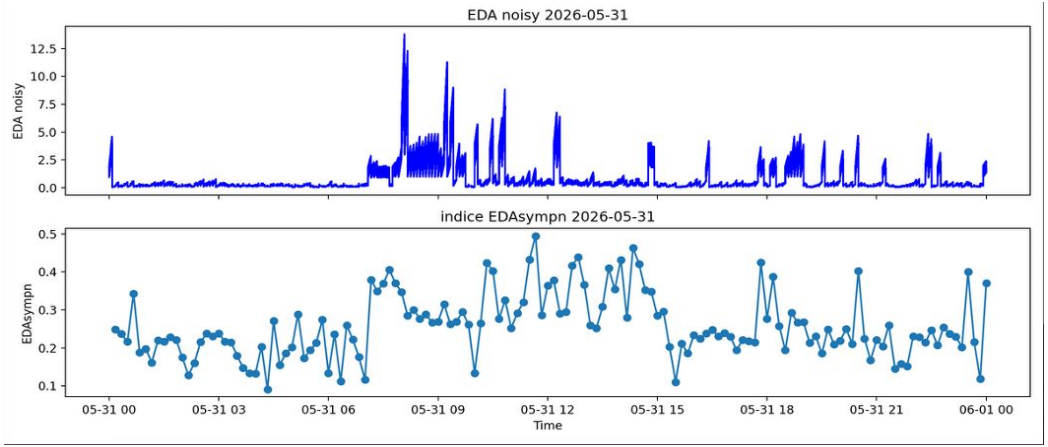
n.u: normalized units. Values are displayed as mean ± standard deviation. *, **, and *** correspond to a difference with Baseline condition with a $p < 0.05$, $p < 0.01$, or $p < 0.001$, respectively. #, ##, and ### correspond to a difference with EMOT condition with a $p < 0.05$, $p < 0.01$, or $p < 0.001$, respectively.

3.2.2. Heart Rate Variability (HRV)

As a main observation, not all HRV indices highlighted a specific response to COG; only HF-HRV markers showed a significative difference between COG and EMOT (Table 2). More explicitly, HF-HRV indices obtained by using VFCDM and WPT3 methods (HF_{VFCDM} and HF_{WPT3}) showed lower values during COG, while RMSSD and DFT-derived indices that are also employed classically to describe the vagal behavior did not reach statistical significance (Table 2).



https://impact.dei.unipd.it/bwthw/data/v1/steps/patients/Jpefaq6m58//day/2026-05-31/



2026-05-08_EDAsympn.json

{
"time":"2026-05-08T00:39:59.995",
"value":0.2391269896
}

Personas

Mary a student at university,

Behaviours: spends a lot of time studying during the week. She often walks around the city either to run errands or to release tension after a study session.

Motivation: she is currently using an app to measure stress that exclusively uses HR continuously throughout the day. During exam sessions she would want an app that tracks mental stress without her everyday physical activity influencing the measurement as much.

Ability inhibitors: difficulty in recognizing when prolonged mental effort and stress are reducing her ability to study

Simon A young Data Analyst,

Behaviours: Sedentary job, spends most of the working day in front of a screen, with limited physical activity during the working week.

Motivation: after a few months at the job he is feeling stressful. He has a wearable device that measures sleep quality and heart rate but , he wants something more sensitive when he works in front of the screen to see his cognitive load throughout the day. Track his mental stress and when the week had lots of stressful days maybe getting more sleep.

Ability Inhibitors: Difficulty recognizing when mental stress is negatively affecting his work performance and recovery.

The Stress Index

$$SI(h) = F_{Vuln,sleep} \cdot \left(HR_{elevation\%}(h) \cdot W_{HR} + EDAsympn_{norm}(h) \cdot W_{EDA} \right)$$

- $HR_{elevation\%}$ is the hourly average **Heart Rate elevation percentage** with respect to a resting baseline computed using HR data from sleep hours.
- $EDAsympn_{norm}$ is the hourly average **EDAsympn normalized** with respect to the maximum physiological value of 0.5.

- $F_{Vuln,sleep}$ is a **stress vulnerability factor** obtained from the Sleep Score:
  - ➔ first computing a **Smoothed Sleep Score** as a weighted average of the previous 4 nights score with exponentially decreasing weights;
  - ➔ then the Smoothed Score is mapped to a multiplier: $F_{Vuln,sleep} = 2.9 - (0.02 \cdot SS_{smoothed})$

The Stress Index

$$SI(h) = F_{Vuln,sleep} \cdot \left(HR_{elevation\%}(h) \cdot W_{HR} + EDAsympn_{norm}(h) \cdot W_{EDA} \right)$$

$$SI(h) = \text{Physical Stress}(h) + \text{Mental Stress}(h)$$



Stress Ring

- **Adaptive redistribution of formula weights** if HR or EDA data are missing: if one sensor fails, 100% of the weight automatically shifts to the available signal, allowing calculation of Stress Index without crashing.

- A **Daily Stress Index** is computed as a weighted average of previous hourly stress indices, where values >60 are assigned double weight (acute stress penalty).

Daily Average: 44/100

The Development Journey: Milestones & Challenges

- Transition in EDA measure from basic EDA peak count to the more robust EDAsympn index, generation of simulated EDA trace with true physiological coherence using **step data as a baseline** and adding controlled random components

- Mixing live HR/steps with a pre-simulated EDA trace would destroy the physiological correlation between the signals. We prioritized **physiological integrity over live tracking**: for the purpose of this demo, we intentionally anchored the app to a specific past date, ensuring all displayed data is synchronized and coherent

- Physical exercise triggers cardiovascular and sudomotor responses (high HR and EDA) that mathematically mimic mental stress. Utilizing step traces and setting a threshold of 1,500 steps per hour, we effectively **exclude periods of physical exertion** from the psychological stress calculation.

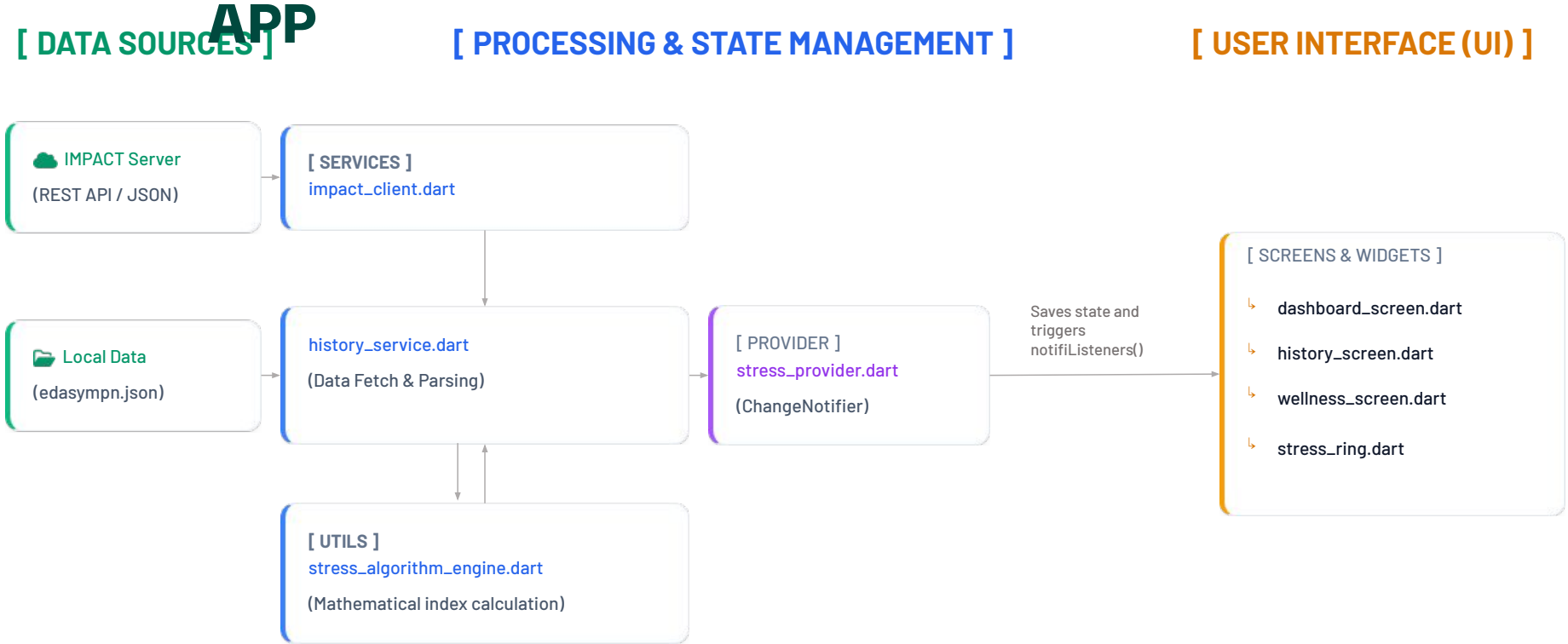
- The initial architecture triggered redundant IMPACT API requests every time a screen was rendered or navigated. We implemented a **local caching**: data is fetched once and injected into the UI via the Provider

- Final development phase dedicated to user experience improvement: **refinement of charts and widgets** to ensure easy-to-read presentation of health data

Team Organization

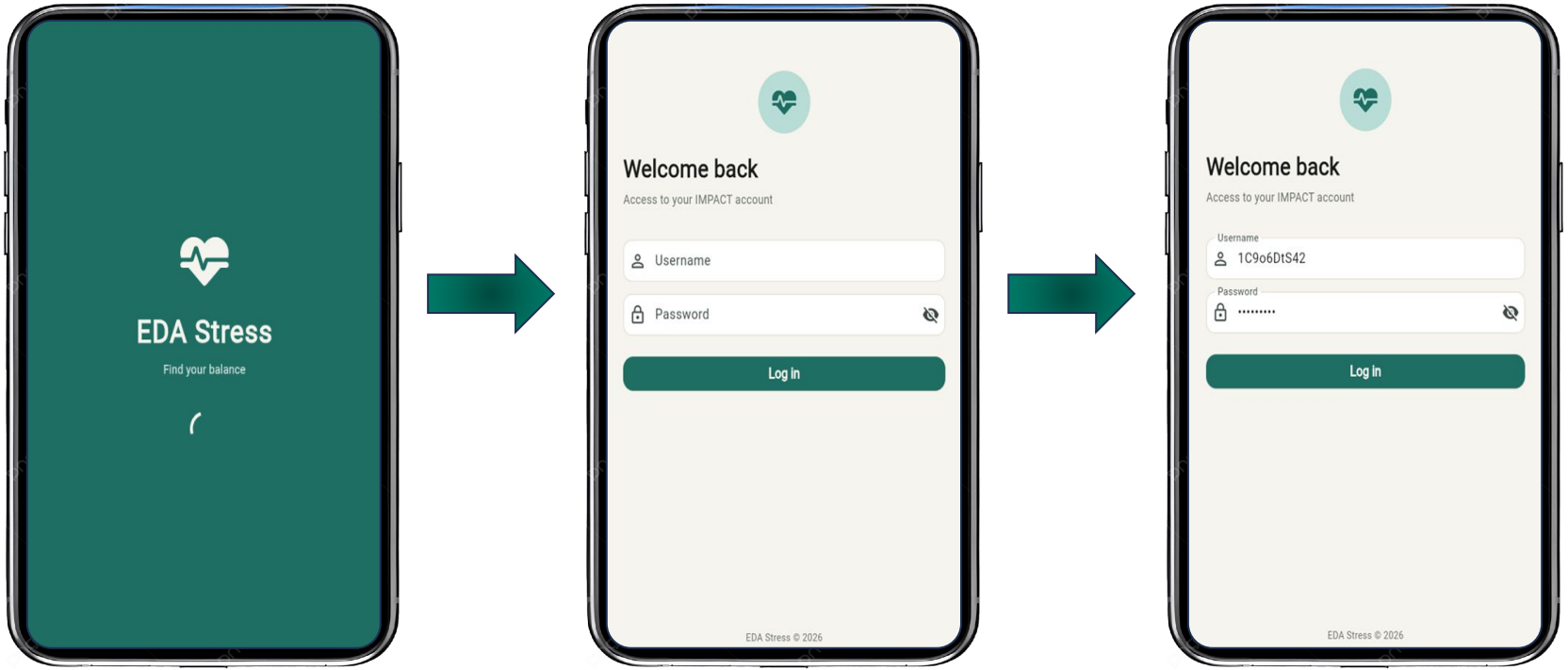
- We worked together as a team: all decisions were discussed and approved democratically
- While everyone contributed globally to the entire code, we maintained specific focuses:
  - ❏ **Terese:** Core idea of a stress monitoring App based on EDA, EDAsympn paper research, and data simulation
  - ❏ **Samuele:** Stress index algorithm design, metrics, and hourly/daily calculation logic
  - ❏ **Riccardo:** UI/UX development, and data management optimization (Provider)
- Code collaboration and version control was achieved using GitHub
- Held regular meetings for brainstorming, progress tracking, and task distribution.

DATA WORKFLOW OF EDA STRESS APP



LOGIN SCREEN

The login screen has a simple design. Users can enter their username and password, and they can choose to show or hide the password. When the login button is pressed, a loading indicator is shown while the app processes the request. If the login is not successful, an error message appears to inform the user.



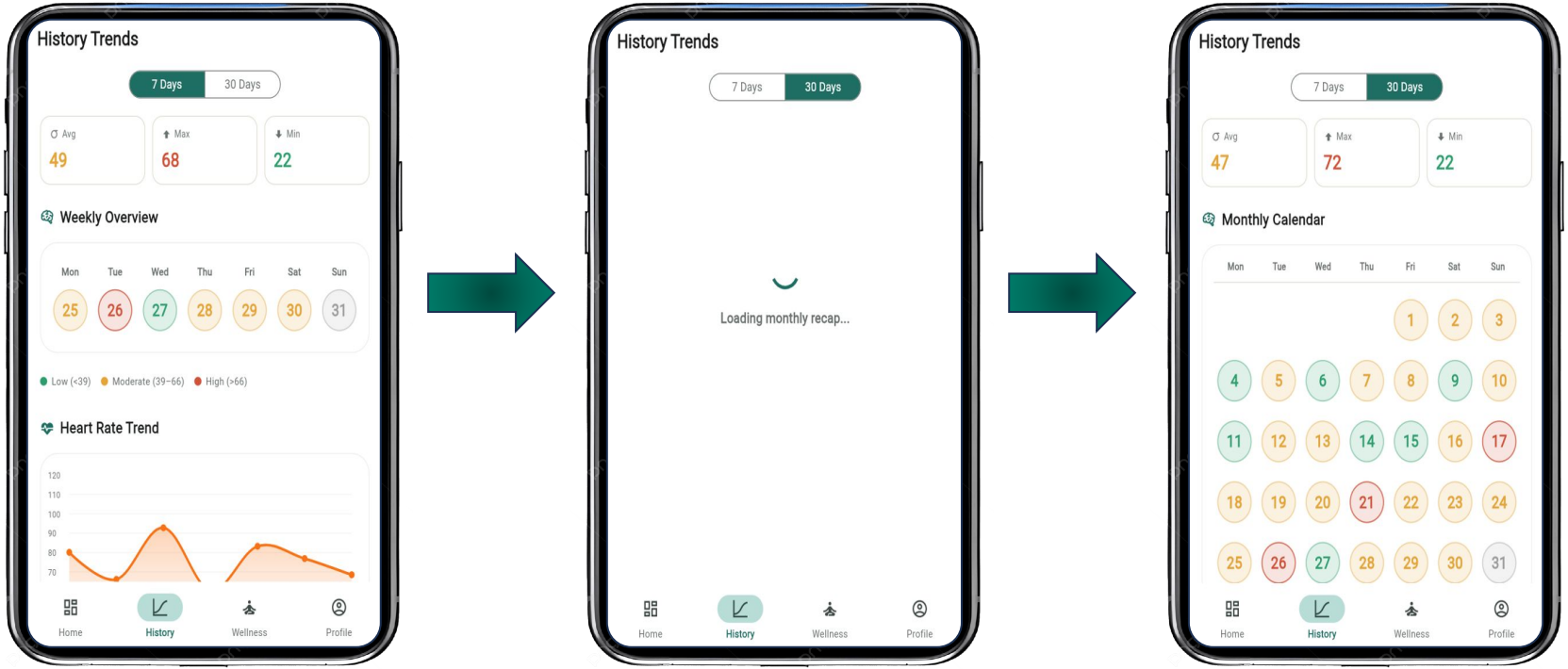
DASHBOARD SCREEN

The Dashboard gives the user a overview of their daily stress and health data. It shows a greeting based on the time of day, the current stress level through the stress ring, and a graph that shows stress and activity during the day. Important information such as sleep, steps, heart rate, and activity is also displayed in simple buttons. By tapping the stress ring, the user can access the Stress Detail screen, where they can better understand their stress level. This screen explains the main factors that affect stress and shows two graphs for physical and mental load, giving the user a clearer view of their daily stress.



HISTORY SCREEN

The History screen allows users to view their stress data over a longer period. They can switch between 7 days and 30 days to easily compare different periods. The screen shows the average, highest, and lowest stress levels, using different colors to make the information easier to understand. Stress data is also displayed in a simple daily view or a calendar style view. Finally, graphs show changes in heart rate and sleep, helping users understand their health trends over time.



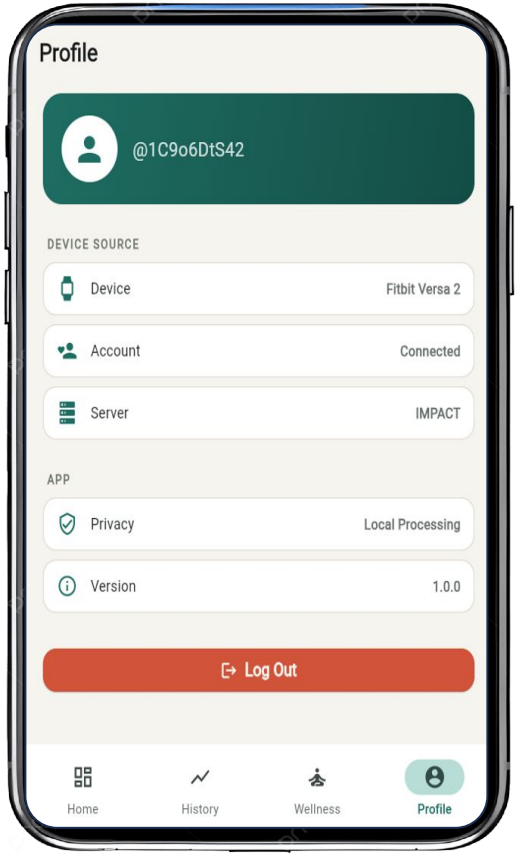
WELLNESS SCREEN

The Wellness screen focuses on helping users improve their well-being and recover from stress. It offers a guided breathing exercise with a simple animation, making it easy to follow. The screen also shows information about sleep, physical activity, and daily progress, helping users understand their recovery. Finally, personalized tips and suggestions are provided based on the user's stress level, giving practical ideas to improve their well-being.



PROFILE SCREEN

The Profile screen brings together the user's personal information, settings, and privacy options. At the top, a green header shows the user's name and profile picture. Below, information about connected devices, the IMPACT server, and privacy is organized in a clear way. At the bottom, the Logout button is highlighted in red to make the action easy to recognize.



Future Developments: From Demo to Market Product

True Real-Time Monitoring

- Integration with wearables (e.g., smartwatches) with continuous EDA and HR sensors
- Transitioning from static, pre-recorded hourly data blocks to dynamic, minute-by-minute live stress tracking

Active Feedback

- Evolving the breathing exercise from a standalone tool into a closed-loop biofeedback system fot the user
- Users will visually see their physiological metrics (HR/EDA and Stress Index) actively decreasing in real-time on their wearable device while performing the exercise.



